The State-Dependent Phillips Curve:
Evidence from Machine Learning

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Overview

Motivation

1. **Pre-2008:** the inflation-unemployment relationship was weak
2. **2008–09:** deep recession, models predict deflation $\rightarrow$ doesn't happen
3. **2021–22:** tight labour markets, models predict modest inflation $\rightarrow$ get 9%

Is the lack of evidence caused by nonlinearities and state dependence?

Key insight

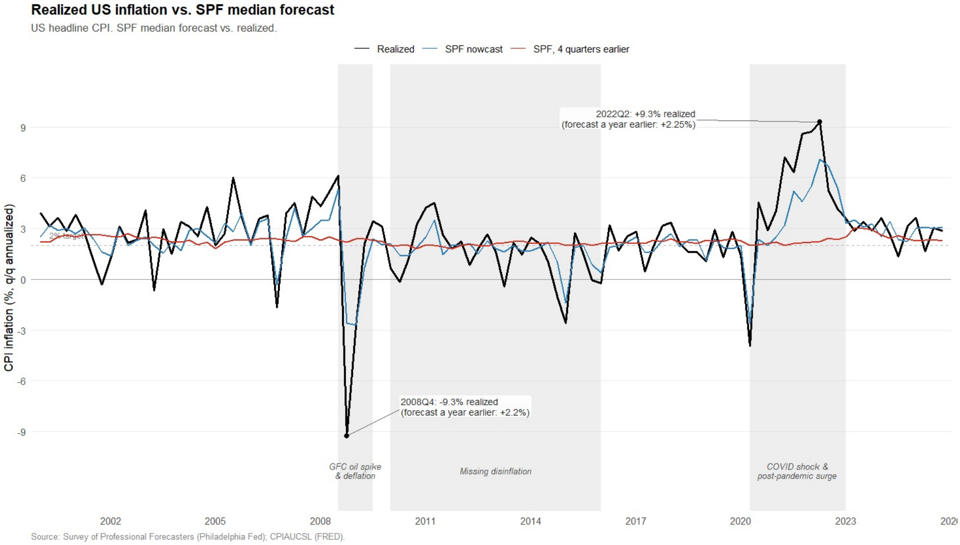
Anchored expectations and mild shocks **masked the true slope**.

Under large shocks the relationship **switches back on** - but different each time

Why it matters

- Inflation forecasts drive **policy rates, forward guidance**, and financial stability assessments
- Systematic forecast errors erode **credibility** and delay policy response

Forecast errors during large shocks



What the Literature Says

Camp 1: Nothing is broken

The curve *looked* flat because stable monetary policy kept expectations anchored — not because the slope changed.

Jørgensen & Lansing (2025)

Early post-pandemic inflation was driven by goods price shocks; no evidence for PC nonlinearity.

Bernanke & Blanchard (2023)

Camp 2: The slope kinks

L-shaped PC — flat when $v/u < 1$, steep when vacancies exceed unemployment.

Benigno & Eggertsson (2023)

Nonlinear terms (u^2, u^3) capture convexity during large tightening episodes.

Ball, Leigh & Mishra (2022)

Camp 3: Regime-dependent

Labour tightness alone is not enough.

The slope shifts with supply-chain stress, expectations anchoring, and shock type — the driver changes by episode.

CBO (2023); Coulombe (2025)

Research Question & Hypothesis

Research Question

Do inflation dynamics exhibit non-linearities and state-dependency?

(Do ML models outperform linear Phillips Curve specifications *specifically during shock episodes*?)

How much of the improvement in forecast performance can be attributed to nonlinearity versus regime-dependent variable selection?

Hypothesis

Linear models fail during shocks because they cannot adapt to **regime changes**. ML models detect when the slope changes and which drivers become dominant.

Relative forecast gains $\equiv$ **empirical diagnostic of nonlinearity and state-dependency**.

Key Concept

State-Dependency

The Phillips Curve slope varies with the economic regime - identical shocks produce different inflation responses depending on slack, inflation level, and supply conditions.

Harding, Lindé & Trabandt (2023); Coulombe (2024)

Contribution

What this paper adds:

- Uses ML as an **agnostic diagnostic tool**
  - **agnostic**: we do not assume any shape
- Links forecast performance to the **nonlinearity** and **state-dependent inflation drivers** debate
  - focus on the **shock periods**
- Disentangles the contributions of non-linearity and variable-selection

Lucas Critique (1976)

Estimated parameters are **not structural constants** — they reflect agent behaviour under a specific regime. When the regime shifts, expectations and price-setting rules shift with it.

The forecasting trap:

- A *pure forecasting* exercise uses historically learned patterns to predict the future — exactly what Lucas warned against
- When the regime changes, **expectations shift**, variables behave differently, and the old parameters become invalid
- 2008 and 2021 are not bad luck — they are the predictable cost of **ignoring state-dependency**

Our position:

- Machine learning methods can incorporate regime changes on a more flexible basis, creating different nodules depending on the nature of the shock. They can capture the changing regime structures, making them more immune to the Lucas critique.

Data and Empirical Strategy

Data

- FRED-MD: 130+ monthly macro series
- Two shock windows:
  - 2008–09 (Great Recession)
  - 2020–22 (COVID surge)
- Rolling-window out-of-sample forecasting

Linear Benchmarks

- AR(4), RW
- Linear Phillips Curve (classic + extensions)

Linearity + Variable Selection

- LASSO / Ridge

Phillips Curve + Nonlinearity

- Decision Trees with PC variables

Non-Linearity + Variable Selection

- Random Forest
- Local Linear Forest

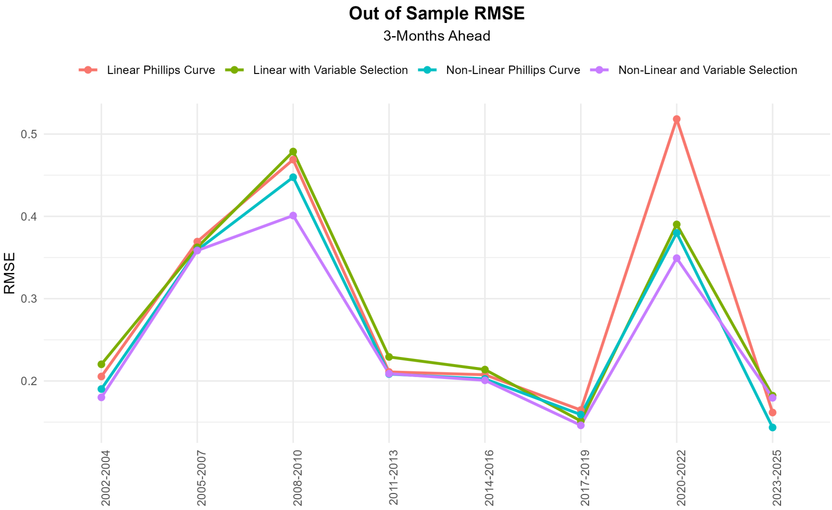
What we compare

- Forecast MSE: ML vs. linear benchmarks
- Is the gap larger during shock periods?
- **Disentangling** the contribution of **non-linearity vs. variable selection**
- Variable importance: what does ML upweight during shocks?

Figure 1

Shock Periods RMSE 3-Step Ahead Out of Sample				
Year	Linear Phillips Curve	Linear with Variable Selection	Non-Linear Phillips Curve	Non-Linear and Variable Selection
2002-2004	0.205	0.218	0.190	0.180
2005-2007	0.369	0.362	0.359	0.358
2008-2010	0.469	0.480	0.448	0.401
2011-2013	0.211	0.227	0.208	0.209
2014-2016	0.208	0.214	0.202	0.201
2017-2019	0.165	0.152	0.159	0.146
2020-2022	0.518	0.390	0.380	0.349
2023-2025	0.162	0.178	0.143	0.180
Average All	0.286	0.276	0.266	0.257

Figure 2


Out of Sample RMSE
3-Months Ahead
Linear Phillips Curve
Linear with Variable Selection
Non-Linear Phillips Curve
Non-Linear and Variable Selection
RMSE
0.5
0.4
0.3
0.2
2002-2004
2005-2007
2008-2010
2011-2013
2014-2016
2017-2019
2020-2022
2023-2025

Ensemble Forecasting

Why combine forecasts?

- No single model dominates across regimes
- Averaging reduces variance and model-specific error
- Weights reveal **which models matter most**

Simple Average: Best 5

Equal-weighted average of the 5 models with lowest out-of-sample RMSE.

Constrained Regression Weights

Regress **realized inflation** on **model forecasts**:

$$\pi_t = \sum_i w_i \hat{\pi}_{i,t} + \varepsilon_t$$

Estimated pre-2011, with $\sum_i w_i = 1,\ w_i \geq 0$.

Larger $w_i \Rightarrow$ model i more informative.

Constrained Ridge Weights

Add ℓ_2 penalty $(\lambda = 5)$ to **shrink weights** toward equality.

Figure 3 — Ensemble Forecasting

Out of Sample RMSE

3-Step Ahead

Year	LASSO	Ridge	ElNet	RW	RSM	AR	RF	LLF	LASSO_L	Ridge_L	ElNet_L	RF_L	LLF_L	best_5	const_1	const_2
2002-2004	0.233	0.235	0.233	0.267	0.351	0.235	0.232	0.221	0.236	0.271	0.236	0.256	0.278	0.237	0.226	0.242
2005-2007	0.354	0.355	0.355	0.427	0.345	0.311	0.312	0.361	0.310	0.313	0.309	0.337	0.324	0.330	0.349	0.338
2008-2010	0.191	0.282	0.205	0.453	0.336	0.296	0.203	0.266	0.252	0.227	0.250	0.265	0.299	0.184	0.153	0.172
2011-2013	0.241	0.213	0.239	0.281	0.326	0.255	0.219	0.198	0.206	0.217	0.203	0.241	0.225	0.218	0.197	0.221
2014-2016	0.167	0.146	0.166	0.219	0.314	0.192	0.150	0.199	0.116	0.154	0.115	0.130	0.177	0.151	0.182	0.180
2017-2019	0.180	0.181	0.179	0.227	0.305	0.230	0.168	0.209	0.151	0.165	0.150	0.128	0.172	0.166	0.206	0.198
2020-2022	0.405	0.362	0.407	0.543	0.302	0.382	0.310	0.389	0.448	0.432	0.450	0.319	0.364	0.315	0.288	0.300
2023-2025	0.355	0.312	0.354	0.291	0.306	0.295	0.307	0.351	0.291	0.290	0.288	0.265	0.256	0.303	0.341	0.324
Average	2.127	2.087	2.137	2.707	2.585	2.194	1.902	2.193	2.011	2.070	2.001	1.942	2.094	1.904	1.942	1.974

Road to Submission: Next Steps

- **Benchmarking against Institutional Forecasts**
  - Transition from monthly to **quarterly frequency** to allow direct comparison with the **Fed**, other forecast institutions and surveys.
- **Dynamic Variable Importance**
  - Implement **rolling-window variable importance** analysis.
  - Goal: Empirically visualize how inflation drivers rotate (e.g., from expectations to supply chain stress to labor tightness) to prove **state-dependency**.
- **Robustness Checks**
  - Testing sensitivity to **rolling window sizes** and different training horizons.
  - Validation across alternative Phillips Curve specifications (e.g., different measures of slack and inflation expectations).

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Thank you

Comments and suggestions welcome.

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